# Eric Hou

hyweric.github.io ● 510-288-0541 ● hyweric@gmail.com ● github.com/hyweric

**Education**

- Mission San Jose High School, Ohlone College Dual Enrollment
    - Relevant Coursework: Classical Mechanics/Electricity&Magnetism, Multivariable Calculus, AP Physics 1, AP Computer Science A, AP Statistics, AP Chemistry, AP Calculus BC
- SAT: 1550/1600 (800 Math, 750 English)

**Skills**

- Hardware: CAD (Autodesk Fusion 360/Onshape), Manufacturing (3DP, Laser-Cutting, CAM), Electronics
- Software: Python, HTML, CSS, JS, C++, Familiar with Arduino [Tools/Frameworks: Flask, OpenCV, Git, Docker, React]

**Projects (hyweric.github.io/projects)**

- Research Paper Accepted@IRC IEEE 2024: Climbing Robot with Adaptive Suction and Suspension Foot Design for Infrastructure Monitoring
- Autodesk Fusion 360 Wire Generator (1000+ Users)
- Optical Flow Egomotion Localization Algorithm using Computer Vision
- Flask Applications: Inverse Kinematics Simulator, PortfolioPro, Stick OpenCV Animator
- 3 Chrome Extensions: Mindful Website Blocker, Distraction Free Youtube, LLM Remark Website Deterrent
- 6502 Computer Made from Scratch

**FTC Team 13356 Hardware Co-Captain (hyweric.github.io/port/ftc/ftc)**

- Accomplishments
    - 2023 and 2024 Northern California Regionals Finalist Alliance Captain: OPR California #3, Consistently T30
    - Qualified for Maryland Tech Invitational (International Competition of World's Top 40 Teams) over Past 3 Years
- Experience
    - Spent 15-30 hours a week designing, manufacturing, testing/collecting data, and integrating HW with SW
    - Things I've Designed (oldest - recent): Extendable Carousel Spinner w/ custom 3DP rails, Suspension Mecanum Drivetrain, Flipping Intake, Dual-Motor Counter Torque Spring Linear Slide Powering Assembly, Stack Intake, Pixel Storage Magazine, Plane Launcher
- Outreach:
    - Designed and manufactured STEM Kits for Rural Communities in Lesotho and underresourced Organizations.
    - Developed Curriculum and taught CAD and programming to Afghan refugees at the Afghan Coalition.
    - Hosted Activities at Tri-Valley Innovation Fair, Santa Clara County Fair, ACSEF, Reid Hillview Airport, Quest Science, Afghan Coalition Health Fair, etc

**Head of Computer Science & Member of Board of Directors - Learnify+ Non-Profit Tutoring Organization**

- Volunteered over 450 hours creating plans and organizing classes for 50+ tutors and 200+ students
- Led the Mission International initiative, expanding the organization's reach to 10 states and 4 countries
- Prepared curriculum for Java, Python, and HTML for other tutors within CS division

**USA International Wushu Team**

- Trained 3x a week (~10 hrs) in Elite Kung Fu Traditional Team to represent the US in World Kungfu Championships in China (Ranked 8 and 18 in 2 categories); Pan-American Wushu (Gold), Golden State Wushu Championship (Gold)

**Horizon AI Intern**

- Ongoing Assignment: SEO-optimized website directory framework project w/ device-responsive layout and filters

**Student Researcher | ASDRP**

- Deep Learning Project - Using quantum annealing/optimization and a deep learning approach to isolate jets from images of particle collision with Dr. McMahan

**Other:** Toshiba ExploraVision Project Honorable Mention (T10%), SASA STEM Advocacy, Golden State Youth Orchestra Violinist, MSJ Robotics Club Officer, School Badminton Team